We note that the first and second figures in Illustration 1.2 represent the functions $f(x) = 2x$ and $f(x) = x^2$ respectively. Usually the variable x is treated as independent variable and y as a dependent variable. The variable x is called the ***argument*** and $f(x)$ is called the value.

(c) Analytical Representation of a Function

If the functional relation $y = f(x)$ is such that f denotes an analytical expression, we say that the function y of x is represented or defined analytically. Some examples of analytical expressions are

$$x^3 + 5, \ \frac{\sin x + \cos x}{x^2 + 1}, \ \log x + 5\sqrt{x}.$$

That is, a series of symbols denoting certain mathematical operations that are performed in a definite sequence on numbers, letters which designate constants or variable quantities.

Examples of functions defined analytically are

$$\text{(i)} \ y = \frac{x-1}{x+1} \quad \text{(ii)} \ y = \sqrt{9 - x^2} \quad (iii) \ y = \sin x + \cos x \quad \text{(iv)} \ A = \pi r^2.$$

One of the usages of writing functions analytically is finding domains naturally. That is, the set of values of x for which the analytical expressions on the right-hand side has a definite value is the natural domain of definition of a function represented analytically.

Thus, the natural domain of the function,

(i)	$y = x^3 + 3$	is	$(-\infty, \infty)$	(ii)	$y = x^4 - 2$	is	$(-\infty, \infty)$
(iii)	$y = \frac{x-1}{x+1}$	is	$\mathbb{R} - \{-1\}$	(iv)	$y = +\sqrt{4 - x^2}$	is	$-2 \le x \le 2$.

Now recall the domain of the functions (i) $y = 2x$, (ii) $y = x^2$, (iii) $y = +\sqrt{x}$, (iv) $y = -\sqrt{x}$ which are analytical in nature described earlier.

Sometimes we may come across piece-wise defined functions. For example, consider the function $f : \mathbb{R} \to \mathbb{R}$ defined as

$$f(x) = \begin{cases} 0 & \text{if} \quad -\infty < x \le -2 \\ 2x & \text{if} \quad -2 < x \le 3 \\ x^2 & \text{if} \quad 3 < x < \infty \end{cases}$$

Depending upon the value of x, we have to select the formula to be used to find the value of f at any point x. To find the value of f at any real number, first we have to find to which interval x belongs to; then using the corresponding formula we can find the value of f at that point. To find $f(6)$ we know $3 \le 6 < \infty$ (or $6 \in [3, \infty)$); so we use the formula $f(x) = x^2$ and find $f(6) = 36$. Similarly $f(-1) = -2, f(-5) = 0$ and so on.

If the function is defined from $\mathbb{R}$ or a subset of $\mathbb{R}$ then we can draw the graph of the function. For example, if $f : [0, 4] \to \mathbb{R}$ is defined by $f(x) = \frac{x}{2} + 1$, then we can plot the points $(x, \frac{x}{2} + 1)$ for all $x \in [0, 4]$. Then we will get a straight line segment joining $(0, 1)$ and $(4, 3)$. (See Figure 1.13)

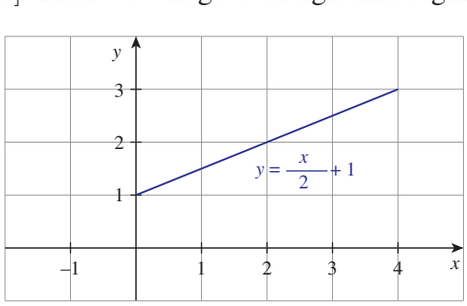


Figure 1.13

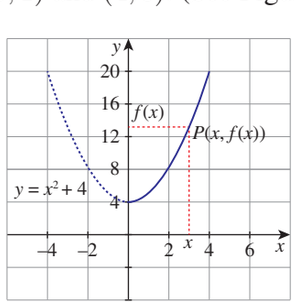


Figure 1.14